

1. Using Bayes' Theorem for Statistical Inference

Question 1

For the given data, the Likelihood function becomes:

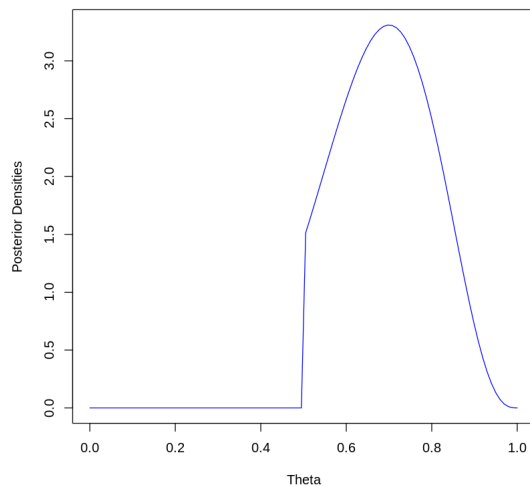
$$L(\theta) = \frac{1408}{227} {}^{10}C_7 \theta^7 (1 - \theta)^3 p(\theta),$$

where $p(\theta)$ is the prior distribution.

- $\theta = 0.75$
 $L(0.75) = \frac{1408}{227} {}^{10}C_7 (0.75)^7 (1 - 0.75)^3 \approx 3.105$
- $\theta = 0.25$
 $L(0.25) = \frac{1408}{227} {}^{10}C_7 (0.25)^7 (1 - 0.25)^3 = 0$
- $\theta = 1$
 $L(0.75) = \frac{1408}{227} {}^{10}C_7 (1)^7 (1 - 1)^3 = 0$

Question 2

```
1 prior <- function(t){
2   ifelse(t >= 0.5 & t <= 1,2,0)
3 }
4 lkl <- function(t){
5   dbinom(7, size=10, t)
6 }
7 marginal <- (227/1408)
8
9 posterior <- function(t){
10   (lkl(t)*prior(t))/marginal
11 }
12
13 t <- seq(0, 1, length.out = 100)
14 posterior_vals <- posterior(t)
15
16 plot(x=t, y=posterior_vals, type='l', col='blue', xlab = 'Theta', ylab = 'Posterior Densities')
```



Question 3

Upon differentiating the Likelihood function, we get:

$$L(\theta) = \frac{1408}{227} {}^{10}C_7 [7(\theta)^6 (1 - \theta)^3 - 3(\theta)^7 (1 - \theta)^2] p(\theta)$$

Note that $p(\theta)$ does not get differentiated because the values it takes are constants.

Equating the above equation to 0, we get

$$7(\theta)^6 (1 - \theta)^3 = 3(\theta)^7 (1 - \theta)^2$$

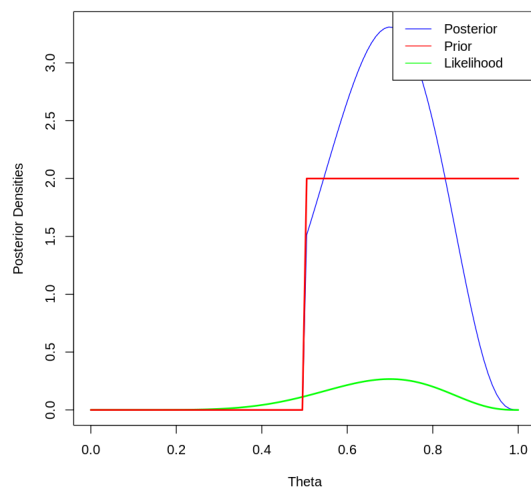
$$\Rightarrow \theta = 0.7$$

The value $\theta = 0.7$ has the maximum posterior density.

Question 4

Here are all three graphs. We can observe that Posterior density looks like the product of Prior distributions and Likelihood functions. The values on the Y-axes of the graphs also shows how the values changed due to the marginal likelihood(in the posterior distribution).

```
1 prior_vals <- prior(t)
2 lkl_vals <- lkl(t)
3 plot(x=t, y=posterior_vals, type='l', col='blue', xlab = 'Theta', ylab = 'Posterior Densities')
4 lines(t, lkl_vals, lwd = 2, col = "green")
5 lines(t, prior_vals, lwd = 2, col = "red")
6
7 legend("topright",
8       legend = c("Posterior", "Prior", "Likelihood"),
9       col     = c("blue", "red", "green"),
10      lwd     = 1)
```



2. Model Building in the Bayesian Framework

Question 1

```
1 dat <- read.table("/content/recognition_c946fb72-19dc-49d2-80b9-ca1a4f1063d7 (1).csv", sep=",", header=T)[,-1]
```

```
1 mu_vals <- seq(200,400,length.out=1000)
2 sigma <- 60
```

```
1 unnorm_post_null <- sapply(mu_vals, function(mu) {
2   lkl_Tw <- prod(dnorm(dat$Tw, mean = mu, sd = sigma))
3   lkl_Tnw <- prod(dnorm(dat$Tnw, mean = mu, sd = sigma))
4   prior_mu <- dnorm(mu, mean = 300, sd = 50)
5
6   return(lkl_Tw * lkl_Tnw * prior_mu)
7 })
```

```
1 plot(mu_vals, unnorm_post_null, type = "l", col = "blue", lwd = 2,
2      main = "1. Unnormalized Posterior of mu (Null Model)",
3      xlab = "mu", ylab = "Unnormalized Density")
```

1. Unnormalized Posterior of mu (Null Model)

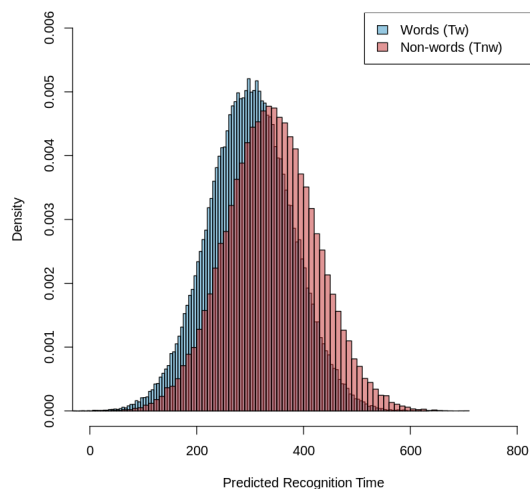
Question 2

```
1 install.packages("truncnorm")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
1 library(truncnorm)
2
3 n <- 100000
4 sigma <- 60
5
6 mu_prior <- rnorm(n, mean = 300, sd = 50)
7 delta_prior <- rtruncnorm(n, a = 0, b = Inf, mean = 0, sd = 50)
8
9 Tw_pred <- rnorm(n, mean = mu_prior, sd = sigma)
10
11 Tnw_pred <- rnorm(n, mean = (mu_prior + delta_prior), sd = sigma)
12
13 hist(Tw_pred,
14      col = rgb(0.2, 0.6, 0.8, 0.5),
15      xlim = c(0, 800),
16      ylim = c(0, 0.006),
17      probability = TRUE,
18      breaks = 100,
19      main = "Prior Predictions(Lexical-Access Model",
20      xlab = "Predicted Recognition Time")
21
22 hist(Tnw_pred,
23      col = rgb(0.8, 0.2, 0.2, 0.5),
24      probability = TRUE,
25      breaks = 100,
26      add = TRUE)
27
28 legend("topright",
29      legend = c("Words (Tw)", "Non-words (Tnw)"),
30      fill = c(rgb(0.2, 0.6, 0.8, 0.5), rgb(0.8, 0.2, 0.2, 0.5)))
```

Prior Predictions(Lexical-Access Model)



Question 3

In the Null Hypothesis model, the parameter δ is fixed exactly at 0. So effectively the distributions from which T_w and $T_n w$ are generated are the same since both are normally distributed with mean μ and variance σ . Hence, the prior predictions for both completely overlap.

In the Lexical-Access Model, the model assumes that delta is drawn from a truncated normal distribution, where the values must be positive. T_w is still drawn from $N(\mu, \sigma)$ but $T_n w$ is now drawn from $N(\mu + \delta, \sigma)$. Because delta adds a strictly positive amount of time, the prior distribution for non-words shifts to the right (indicating longer recognition times). The prior distribution of $T_n w$ is also slightly wider due to the added uncertainty of δ .

Question 4

The observed data clearly shows that non-words take noticeably longer to recognize than words. The Null model assumes there is absolutely no difference between the two ($\delta=0$), meaning its predictions completely overlap. On the other hand, the Lexical-access model adds a strictly positive time delay δ for non-words, natively predicting that they will take longer to recognize. Because this built-in shift perfectly matches the pattern seen in the real data, the Lexical-access model is much more consistent with reality.

Question 5

```
1 dat <- read.table("/content/recognition_c946fb72-19dc-49d2-80b9-ca1a4f1063d7 (1).csv", sep=",", header=T)
2 Tw <- dat$Tw
3 Tnw <- dat$Tnw
4 sigma <- 60
5
6 delta_vals <- seq(0, 150, length.out = 100)
7 mu_vals <- seq(200, 400, length.out = 100)
8
9 unnorm_post_delta <- numeric(length(delta_vals))
10
11 for(i in seq_along(delta_vals)) {
12   delta <- delta_vals[i]
13
14   prior_delta <- dtruncnorm(delta, a = 0, b = Inf, mean = 0, sd = 50)
15
16   marginal <- 0
17
18   for(j in seq_along(mu_vals)) {
19     mu <- mu_vals[j]
20
21     lik_Tw <- prod(dnorm(Tw, mean = mu, sd = sigma))
22     lik_Tnw <- prod(dnorm(Tnw, mean = (mu + delta), sd = sigma))
23
24     prior_mu <- dnorm(mu, mean = 300, sd = 50)
25
26     marginal <- marginal + (lik_Tw * lik_Tnw * prior_mu)
27   }
28
29   unnorm_post_delta[i] <- prior_delta * marginal
30 }
31
32 plot(delta_vals, unnorm_post_delta, type = "l", col = "purple", lwd = 2,
33       main = "Unnormalized Posterior of delta (Lexical-Access Model)",
34       xlab = "delta (Additional Time)", ylab = "Unnormalized Posterior Density")
```

